

Titel

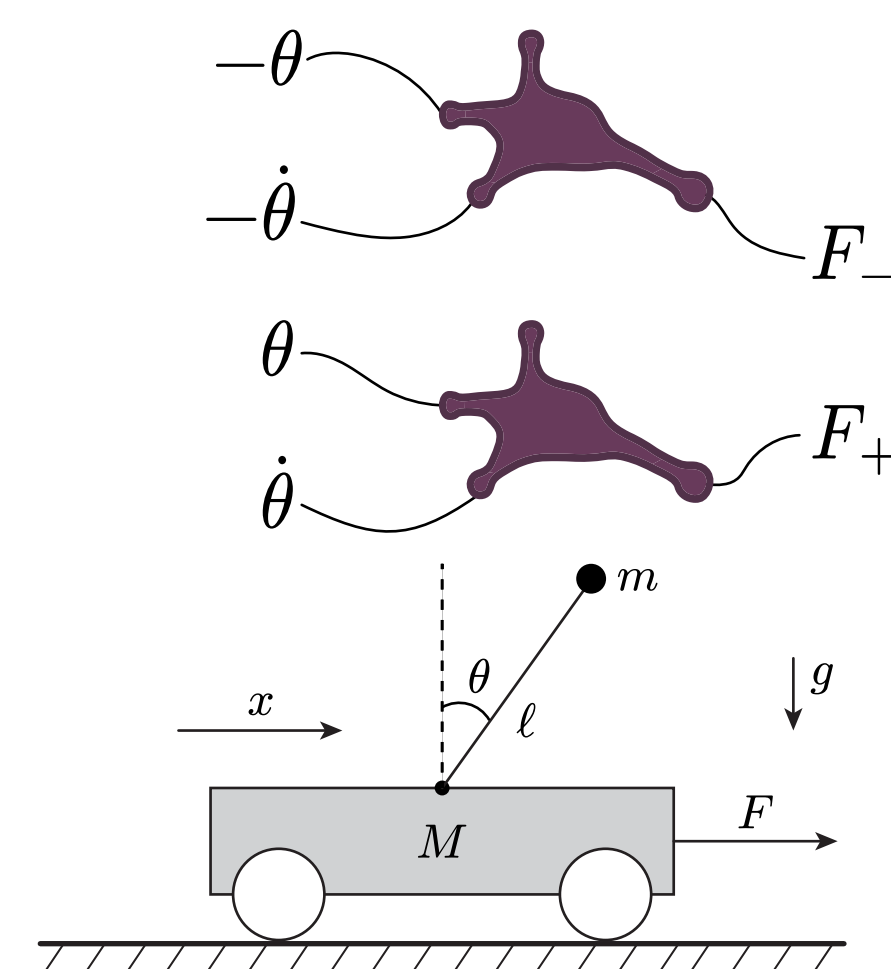
Titel

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Introduction

The leaky integrate-and-fire model offers an energy-efficient method for neuromorphic computing [1][2]. The balancing cartpole is a canonical benchmark for neuromorphic computing. Prior research has addressed cartpoles starting in the upright position using few neurons. However, behaviour of larger networks controlling cartpoles starting in the bottom position is still unknown.

Method

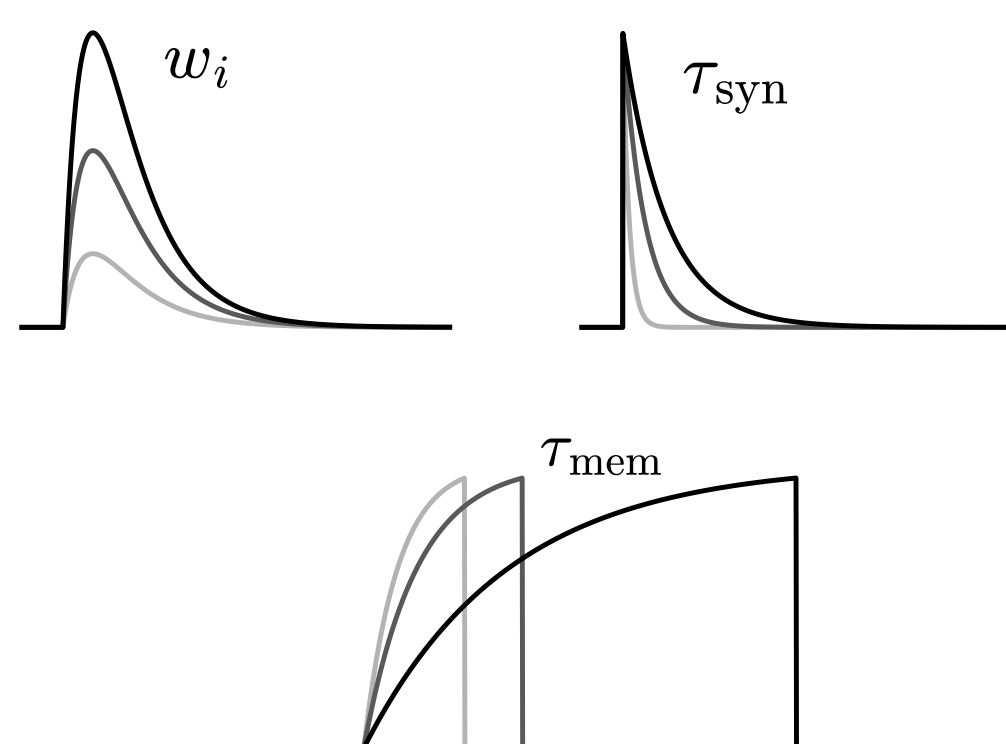


Theory

Lu.i neurons capture the essential neuron dynamics with the leaky integrate-and-fire model which describes how the neuron's membrane potential changes over time. [1]

$$\tau_{\text{mem}} \frac{dV_{\text{mem}}(t)}{dt} = -[V_{\text{mem}}(t) - V_{\text{leak}}] + I_{\text{syn}}(t)/g_{\text{leak}}$$

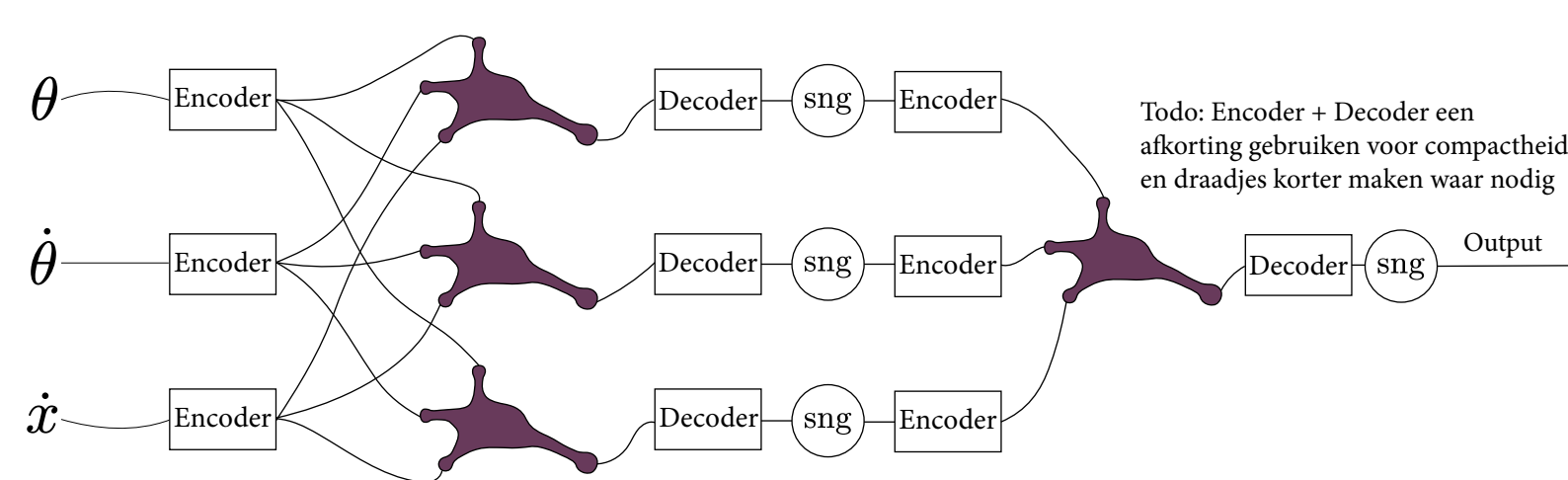
$$I_{\text{syn}}(t) = w_i \exp\left(-\frac{t - t_{\text{spike}}}{\tau_{\text{syn}}}\right) \quad \text{for } t > t_{\text{spike}}$$



Results

$$\ddot{\theta} = \frac{(M + m) g \sin \theta - \cos \theta (F + m \ell \dot{\theta}^2 \sin \theta)}{\ell (M + m \sin^2 \theta)}$$

$$\ddot{x} = \frac{F + m \ell \dot{\theta}^2 \sin \theta - m g \sin \theta \cos \theta}{M + m \sin^2 \theta}$$



Conclusion

[1] Stradmann et al. (2025). *Lu.i – A low-cost electronic neuron for education and outreach*. Trends in Neuroscience and Education, 38, 100248.

[2] Kim et al. (2023). Sharing leaky-integrate-and-fire neurons for memory-efficient spiking neural networks. Frontiers in Neuroscience, 17, 1230002.